WORKED SOLUTION

STEP III 2007, Question 5

Pure • Higher Derivatives and Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q5

The key idea

Everything becomes simple once we write r and $\mathrm{d}\theta/\mathrm{d}x$ in terms of θ alone. From $\coth\theta = x$ we get $r = 1/\sinh\theta$, and then $\mathrm{d}r/\mathrm{d}x = \cosh\theta$ and $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$. Differentiating $\cosh n\theta / r^n$ then produces exactly the compound-angle formula for $\cosh(n+1)\theta$, which sets up an induction.

Solution

Setting up

Since $x > 1$, $\coth\theta = x$ has a unique solution with $\theta > 0$, so $\sinh\theta > 0$. Using $\coth^2\theta - 1 = \operatorname{cosech}^2\theta$,

$$r^2 = x^2 - 1 = \coth^2\theta - 1 = \frac{1}{\sinh^2\theta}, \qquad \text{so} \qquad r = \frac{1}{\sinh\theta},$$

taking the positive root because both r and $\sinh\theta$ are positive.

Two derivatives will do all the work. Differentiating $\coth\theta = x$ with respect to x,

$$-\operatorname{cosech}^2\theta \frac{\mathrm{d}\theta}{\mathrm{d}x} = 1 \quad \Longrightarrow \quad \frac{\mathrm{d}\theta}{\mathrm{d}x} = -\sinh^2\theta = -\frac{1}{r^2}.$$

Differentiating $r^2 = x^2 - 1$ gives $2r\dfrac{\mathrm{d}r}{\mathrm{d}x} = 2x$, so

$$\frac{\mathrm{d}r}{\mathrm{d}x} = \frac{x}{r} = \coth\theta\,\sinh\theta = \cosh\theta.$$

The first two derivatives

Since $y = \ln r^2 = 2\ln r$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{r}\frac{\mathrm{d}r}{\mathrm{d}x} = \frac{2\cosh\theta}{r},$$

as required. For the second derivative it is cleanest to write $1/r = \sinh\theta$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2\cosh\theta\sinh\theta = \sinh 2\theta$. By the chain rule,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 2\cosh 2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}x} = -\frac{2\cosh 2\theta}{r^2},$$

as required.

The third derivative

Now write $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -2\cosh 2\theta\sinh^2\theta$ and use the product rule, with $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$:

$$\begin{aligned}\frac{\mathrm{d}^3y}{\mathrm{d}x^3} &= -2(2\sinh 2\theta\sinh^2\theta + 2\cosh 2\theta\sinh\theta\cosh\theta)\frac{\mathrm{d}\theta}{\mathrm{d}x}\\ &= 4\sinh^3\theta(\cosh 2\theta\cosh\theta + \sinh 2\theta\sinh\theta)\\ &= 4\sinh^3\theta\cosh 3\theta,\end{aligned}$$

by the identity $\cosh(A+B) = \cosh A\cosh B + \sinh A\sinh B$. So

$$\frac{\mathrm{d}^3y}{\mathrm{d}x^3} = \frac{4\cosh 3\theta}{r^3}.$$

The general formula

The pattern $2\cosh\theta/r$, $-2\cosh 2\theta/r^2$, $4\cosh 3\theta/r^3$ suggests the coefficients $2\cdot 0!$, $-2\cdot 1!$, $2\cdot 2!$, so we conjecture

$$\frac{\mathrm{d}^ny}{\mathrm{d}x^n} = (-1)^{n-1}\,\frac{2\,(n-1)!\,\cosh n\theta}{r^n} \qquad (n \geq 1).$$

Ask yourself What is the general version of the calculation we just did for the third derivative?

It is the derivative of $\cosh n\theta/r^n = \cosh n\theta\,\sinh^n\theta$. By the product and chain rules, with $\mathrm{d}\theta/\mathrm{d}x = -\sinh^2\theta$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(\cosh n\theta\,\sinh^n\theta) &= (n\sinh n\theta\,\sinh^n\theta + n\cosh n\theta\,\sinh^{n-1}\theta\cosh\theta)(-\sinh^2\theta)\\ &= -n\sinh^{n+1}\theta\,(\cosh n\theta\cosh\theta + \sinh n\theta\sinh\theta)\\ &= -n\sinh^{n+1}\theta\,\cosh(n+1)\theta,\end{aligned}$$

that is,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh n\theta}{r^n}\right) = -\frac{n\cosh(n+1)\theta}{r^{n+1}}.$$

Now the induction. The formula holds for $n = 1$ (it is the first derivative found above). Suppose it holds for some $n = k \geq 1$. Then, using the calculation above with $n = k$,

$$\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = (-1)^{k-1}\,2\,(k-1)!\cdot\left(-\frac{k\cosh(k+1)\theta}{r^{k+1}}\right) = (-1)^k\,\frac{2\,k!\,\cosh(k+1)\theta}{r^{k+1}},$$

which is the formula for $n = k + 1$. By induction,

$$\boxed{\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = (-1)^{n-1}\,\frac{2\,(n-1)!\,\cosh n\theta}{r^n} \quad \text{for all } n \geq 1.}$$

(The cases $n = 2$ and $n = 3$ agree with what we found directly.)

A second route

Partial fractions give an independent derivation. Since $y = \ln(x-1) + \ln(x+1)$,

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = (-1)^{n-1}(n-1)!\,[(x-1)^{-n} + (x+1)^{-n}].$$

But $x \pm 1 = \dfrac{\cosh\theta \pm \sinh\theta}{\sinh\theta} = r\,\mathrm{e}^{\pm\theta}$, so the bracket is $r^{-n}(\mathrm{e}^{-n\theta} + \mathrm{e}^{n\theta}) = 2\cosh n\theta / r^n$, the same formula. This also explains where the substitution comes from: θ is half of $\ln\dfrac{x+1}{x-1}$.

Checks

- At $x = 2$: partial fractions give $\mathrm{d}^3y/\mathrm{d}x^3 = 2\left(1 + \frac{1}{27}\right) = \frac{56}{27} \approx 2.0741$, and $4\cosh 3\theta / r^3$ with $\theta = \tanh^{-1}\frac{1}{2}$, $r = \sqrt{3}$ gives the same value.
- We checked the first three derivatives against finite differences of $\ln(x^2-1)$ at 200 random points with $x > 1$, and the general formula against the partial-fraction form for $n \leq 14$ (agreement to about 10^{-14}).
- Signs: $\ln(x^2-1)$ is increasing and concave for $x > 1$, matching $y' > 0$ and $y'' < 0$.

Common slips

- Writing $\mathrm{d}\theta/\mathrm{d}x = +\sinh^2\theta$: the derivative of $\coth\theta$ is $-\operatorname{cosech}^2\theta$, and the minus sign is what makes the derivatives alternate.
- Getting the factorial wrong by one: the nth derivative carries $(n-1)!$, since the first derivative has coefficient $2 = 2 \cdot 0!$.
- Checking a few cases and stopping: "with proof" needs the general inductive step.

Beyond STEP

The neat closed form for every derivative is a glimpse of complex analysis: $\ln(x^2-1)$ has singularities at $x = \pm 1$, and its higher derivatives are governed by the distances to them, here $x \mp 1 = r\mathrm{e}^{\mp\theta}$. In a first-year analysis course, formulae like this are what you need to bound the remainder in a Taylor series and find its radius of convergence.

Needham's *Visual Complex Analysis* explains beautifully why singularities control power series.

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